Journal of Building Engineering 111 (2025) 113233



Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe





A review of shrinkage and restrained shrinkage cracking in 3D concrete printing

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ARTICLE INFO

Keywords:
3D concrete printing
Drying shrinkage
Plastic shrinkage
Autogenous shrinkage
Restrained shrinkage cracking

ABSTRACT

The benefits related to the use of 3-dimensional concrete printing (3DCP) have been recognized worldwide. Key advantages include cost and time-related reductions in addition to reduced labour and maintenance requirements. However, due to this technology's novelty, many challenges must be addressed before it can be fully integrated into the construction industry. This paper reviews the phenomenon of shrinkage and the consequent shrinkage cracking. Due to the low volume of aggregates, hence high paste and fines content, and early exposure, 3D printed concrete (3DPC) is more likely to experience large shrinkage strains in comparison to cast concrete, to an extent where elements are at risk of cracking. Other factors, like early exposure, excessive plastic shrinkage and curing regimes also contribute to cracking. The reviewed literature shows that reported shrinkage strains of 3DCP elements are significantly larger than those of cast concrete. In addition, environmental and curing conditions, mix proportions, admixtures and specimen sizes have been found to influence the rate and magnitude of shrinkage. Variance in shrinkage strains between studies is partially attributed to the varying open time definition and difference between shrinkage measurement methods. This paper concludes that a standardized testing method should be developed to measure plastic and drying shrinkage for 3DCP elements to enable readers to easily compare the reported shrinkage strains from respective authors and develop a shrinkage database to identify governing factors and design guidelines.

1. Introduction

Due to the benefits and extensive applicability, the potential of concrete printing is recognized worldwide [1]. Since the emergence of 3DCP approximately two decades ago, substantial developments have been made, primarily due to the potential reduction in construction time along with reduced maintenance, labour, operational and material costs associated with this technology [2–4]. Zhang et al. [5] reported a notable reduction in waste, ranging from 30 to 60 %, a 50–80 % decrease in labour costs and a 50–70 % reduction in production time. Moreover, the cost of formwork is avoided with this technology. Consequently, a reduced labour force is required with an absence of formwork [6]. The capability to construct concrete structures without formwork enables the extrusion of complex geometries assisted with computer-aided design (CAD) [7]. Such intricate structures are of particular interest to the automotive, aerospace and medical industries, underscoring the potential use of 3D printing technology beyond traditional construction applications [8].

Although many benefits are associated with 3DCP, many challenges must be addressed before it can be fully industrialized.

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https://doi.org/10.1016/j.jobe.2025.113233
Received 11 March 2025; Received in revised form 28 May 2025; Accepted 17 June 2025
Available online 18 June 2025
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Currently, there are no established regulatory frameworks or guidelines for the design of 3DCP to ensure that these structures meet appropriate limit states of serviceability and safety requirements [4,9]. Another significant drawback of 3DCP compared to cast concrete is the limited incorporation of steel reinforcement in printed structures. Application of this technology is currently limited to predominantly non-structural use. However, potential reinforcing strategies have been explored, investigated and tested in literature, as noted in references [10–12]. The challenge related to the inclusion of reinforcement hinders the large-scale implementation of 3DCP in practice, restricting it primarily to low-rise structures [13].

In comparison to conventionally cast concrete, 3DCP mixes have a sizeably lower aggregate content. The aggregate-to-binder ratios of published 3DCP mixes can be as much as four times lower than those in conventional concrete. Printable mixes typically have a significantly higher fine content to achieve the sought-after fresh-state properties [14]. Aggregate particle sizes in 3DCP mixes are usually smaller than 5 mm, limited by the printer nozzle size or the pump [15,16]. The combination of low aggregate-to-binder ratios and the reduced volume of coarse aggregates relative to that of cast concrete with the lack of curing and formwork, has been found to increase both plastic and drying shrinkage and, in some cases, has led to shrinkage cracking [14,17,18]. For the purposes of this research, the term “shrinkage cracking” is defined as shrinkage-induced cracking, both before and after the final setting time of concrete is reached, whereas “long-term shrinkage cracking” is defined as cracking after 24 h post-printing [19]. Shrinkage presents a significant challenge for 3DCP [20]. Three-dimensional printed concrete (3DPC) is particularly susceptible to shrinkage cracking compared to conventional concrete due to the markedly higher shrinkage strains experienced by printed elements. The high surface-to-volume ratio exhibited by printed members contribute to the excessive shrinkage. This enables significant environmental moisture loss, thereby accelerating the shrinkage rate [14].

Lower aggregate-to-binder ratios result in higher binder volumes present in 3DCP mixes, with ordinary Portland cement (OPC) being used as a significant portion of the binder [21]. Since OPC accounts for roughly 10 % of global emissions, there is a growing trend to adjust mix designs to partially substitute OPC with supplementary cementitious materials (SCM) to improve the sustainability of 3DCP [22,23]. SCMs like fly ash, silica fumes, metakaolin, glass powder and blast furnace slag are commonly incorporated in 3DCP mixes [24–26]. Although SCMs improve the sustainability of 3DCP mixes, the influence of SCMs on shrinkage has been reported to vary depending on the type of SCM used. For instance, silica fumes has been found to increase the risk of shrinkage cracking, while the incorporation of fly ash in mortar mixes decreased drying shrinkage at 90 days [27,28]. This phenomenon is partially attributed to the small particle size of silica fume, which results in a higher surface area and, consequently, a greater water demand, in contrast to fly ash with spherical particles and low water demand. Early-age shrinkage cracking, referred to as plastic shrinkage cracking (PSC), is defined in this study as shrinkage-related cracking that occurs within the first 24 h post-printing. However, another study found that Furthermore, earth-based mixes are also developed from various waste sources to address the sustainability of 3DCP mixes [29]. To further promote sustainability and circularity, there has also been an effort to substitute natural aggregates with recycled fine and coarse aggregates [30,31].

In addition to the shrinkage-related challenges associated with 3DCP, it is imperative to control the fresh state properties to ensure that extrudability – the ability to extrude concrete without blocking the nozzle [32], pumpability – the ease with which fresh concrete flows to the nozzle without segregation [33], and buildability – the ability of layers to support subsequently extruded layers [34], requirements are all met. These fresh-state properties represent key process-related requirements that must be satisfied while addressing the previously discussed challenges related to 3DCP [16,35]. Specific rheological properties have been proposed to achieve a successful print, where 3DCP mixes should demonstrate a relatively low plastic viscosity, appropriate dynamic and high static yield stress [36]. Plastic viscosity and dynamic yield stress affect the pumpability of a mix, while static yield stress and thixotropy influences the buildability, print speed and shape stability of the concrete mix [37–39].

Excessive shrinkage presents one of the prominent challenges currently experienced with the use of 3DCP. The consequent shrinkage-related cracking, both at an early age and over the long term, undermines the serviceability and durability of additively manufactured structures [14]. Addressing shrinkage-related cracking in printed structures is essential for fully integrating this technology into the construction industry. To address this challenge, it is crucial to mitigate early-age shrinkage, primarily attributed to plastic shrinkage, through effective internal and external curing mechanisms. Also, the importance of certain shrinkage mitigating materials that improve internal curing properties, like construction and demolition waste, light-weight aggregate and superabsorbent polymers cannot be neglected [40,41]. Shrinkage reducing admixtures and expansive agents are also used to mitigate shrinkage [42]. Additionally, long-term shrinkage mechanisms related to drying shrinkage must be considered during the design phase, such as including movement joints to facilitate shrinkage displacements. In the context of this research, “early-age shrinkage” describes shrinkage that occurs within 24 h post-printing, whereas “long-term shrinkage” denotes shrinkage that occurs after 24 h from extrusion.

Consequently, this paper conducts a systematic literature review to report on the published shrinkage strains of the respective shrinkage mechanisms at different ages after extrusion. The first objective is to report the shrinkage magnitude of 3DPC and compare it to cast concrete. Secondly, due to the complexity relating to the phenomenon of shrinkage, this paper investigates and reports on factors that influence shrinkage and contribute to shrinkage cracking. Thirdly, methods of measuring shrinkage are reviewed, and future research needs are identified and prioritized toward solving the shrinkage-related challenges of 3DCP. The last objective is to compare the fundamental mechanisms, magnitudes and influential factors that govern shrinkage in printed and cast concrete.

2. Literature review methodology

This systematic literature review examines shrinkage and restrained shrinkage cracking of primarily extrusion-based concrete, encompassing shrinkage mechanisms, influential factors, reported strains and brief case studies related to restrained shrinkage and

shrinkage cracking. Reviewing the existing literature on cast concrete, the primary aim is to emphasize differences in underlying mechanisms and variances in reported shrinkage strains. A review of the early-age shrinkage measurement methods has been conducted by Kurup et al. [43]. Their review focuses on mechanisms contributing to early-age shrinkage, mitigation measures, and methods for measuring it. Similarly, Zhu et al. [44] reviewed research progress on the shrinkage properties of 3DCP, focusing on the excessive magnitude of shrinkage, shrinkage measuring methods, and factors that influence the rate and magnitude of shrinkage. This review shares certain thematic similarities to those of Kurup et al. [43] and Zhu et al. [44]. However, this review's primary objectives differ from those set out in the respective studies. For instance, this review summarises reported shrinkage values for printed and cast concrete and compares the underlying differences in the respective shrinkage mechanisms between cast and 3DPC. It also compares the differences in shrinkage magnitude between printed and cast concrete.

This review is done in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) framework [45,46]. developed to ensure review rigour and transparency.

2.1. Data collection methodology

A literature search was conducted on the Scopus database on the February 3, 2025 for 3DPC and on February 17, 2025 for cast concrete. Scopus is recognized as a key source of journals to be used in systematic reviews [47,48]. The initial search used the "title, abstract and keywords" function. Boolean operators were used to include documents based on the presence of primary, secondary and tertiary terms as defined in Tables 1 and 2. The document screening parameters, outlined in Table 3, are selected to ensure transparency and avoid author bias [49].

2.2. Inclusion and exclusion of data

2.2.1. Cast concrete

Given the well-established nature of cast concrete, substantial research has been conducted in this field. An initial Scopus database search yielded 9015 documents using the search terms in Table 1. Of these documents, 8368 documents were removed as these documents do not meet the minimum citation threshold. Additionally, 12 non-journal documents and 134 documents that do not include "shrinkage" in the abstract, title or keyword were removed. The abstracts and titles of the remaining 501 documents were manually screened for relevance to this review. Ultimately, 75 documents were selected for inclusion in this shrinkage review.

2.2.2. 3DPC

The initial Scopus database search for 3DPC using the keywords shown in Table 2 found 411 documents. Following the initial search same procedure as for cast concrete and requiring that the abstract had to reference 3DCP shrinkage or additive construction, 80 documents were included in this literature review.

3. Reported shrinkage

Shrinkage poses a significant challenge in concrete, whether it be conventional concrete or 3DPC [20]. However, for 3DPC, shrinkage starts immediately after extrusion, which differs from cast concrete, where shrinkage typically starts after the bleeding and settlement stage of concrete [50]. Shrinkage occurs as a combination of different mechanisms that are influenced by a range of factors of which some include environmental conditions, surface-to-volume ratio, element geometry, binder type, water-to-cement ratio, creep, curing regimes, paste fraction and aggregate volume fraction and restraint combinations [51,52]. This section outlines the respective shrinkage mechanisms and summarises the corresponding strain from relevant literature.

3.1. Shrinkage mechanisms

Respective shrinkage mechanisms occur simultaneously or sequentially, collectively contributing to the total shrinkage strain. The

Table 1
Search parameters used to obtain shrinkage-relevant publications of cast concrete.

Primary terms	Secondary terms
Concrete	Drying shrinkage
	Plastic shrinkage
	Carbonation shrinkage
	Chemical shrinkage
	Thermal shrinkage
	Shrinkage
	Restrained shrinkage
	Shrinkage cracking
	Plastic shrinkage cracking
	Free shrinkage

Table 2
Search phrases used to obtain shrinkage-relevant publications of 3DPC.

Primary terms	Secondary terms	Tertiary terms
3D Printing	Concrete	Drying shrinkage
3D Printers	Mortar	Plastic shrinkage
Additive construction	Cement	Shrinkage
Additive manufacturing	Cementitious materials	Cracking
Digital construction	Review	Shrinkage deformation
Digital manufacturing		Early-age deformation
Rapid prototyping		Plastic shrinkage cracking
Construction Automation		Long-term shrinkage

Table 3
Document screening parameters for 3DPC.

Criteria	Limit	Reason for omission
Source type	Journal	Quality and peer-reviewed
Subject areas	Engineering Material science	Focus area
Language	English	Preferred language
Relevance	Include keyword: shrinkage	Ensuring applicability to review

nature of these mechanisms differs. Some mechanisms, like plastic [53] and drying shrinkage [54] are associated with water loss, while others, like autogenous [55], carbonation [56] and chemical shrinkage [57] arise from chemical reactions within the concrete. In contrast, thermal shrinkage is related to the fluctuations in temperature, either as a result of the hydration reaction, which may cause thermal shrinkage cracking in mass concrete due to differential cooling, or the ambient temperature fluctuations due to day and night temperature fluctuations [58]. These mechanisms also occur at different stages after extrusion, as shown in Fig. 1. It should be noted that for the purpose of this review, "early age" refers to shrinkage that occurs within 24 h after printing and "long-term" refers to shrinkage after 24 h post-printing, as illustrated in Fig. 1 [19].

3.1.1. Plastic shrinkage

The mechanism of plastic shrinkage occurs when concrete is still in the fresh, plastic state before the final setting time of the specific mix and is considered an early-age shrinkage mechanism as illustrated in Fig. 2 [59]. It is arguably the most prominent shrinkage mechanism in 3DPC, which results in adverse side effects, which are discussed in this section. In freshly extruded concrete, water tends to travel upward towards the surface, where it settles as a thin layer of water known as bleeding water, while solid particles tend to settle as a result of their greater density relative to the water [60,61]. The upward propagation of water induces intergranular stresses within the concrete. As the thin layer of bleeding water evaporates from the surface, the concrete element starts to dry. Rapid bleed water evaporation occurs because of the absence of formwork, which in turn results in the formation of negative capillary pressures throughout the pore network as pore water evaporates [61,62]. As a result of a rapid evaporation rate in combination with the inability of the bleed water to replace the evaporated water, plastic shrinkage occurs [43]. This rapid evaporation not only promotes drying and plastic shrinkage, but also hinders the hydration reaction, especially within the first 24–28 h. This leads to an incomplete hydration reaction, particularly in the surface regions, while the cement hydration reaction continues at internal regions, where more favourable conditions are present. Consequently, this results in the following side effects: firstly, a weaker microstructure can be associated with the surface regions due to the lower C-S-H formation and secondly, more stable, fully hydrated inner regions provide internal restraint to shrinkage and surface regions.

Shortly after printing, typically in the first hour or two for normal setting mixes, 3DPC will likely exhibit the highest rate and a large magnitude of total shrinkage as the concrete is most vulnerable to environmental conditions [64]. Therefore, the curing regime closely influences plastic shrinkage. The high quantity of fine aggregates and low volume of coarse aggregates in 3DCP mixes have also promoted plastic shrinkage [63,64]. The large magnitude of early-age shrinkage strains present in fresh 3DCP extrusions leaves

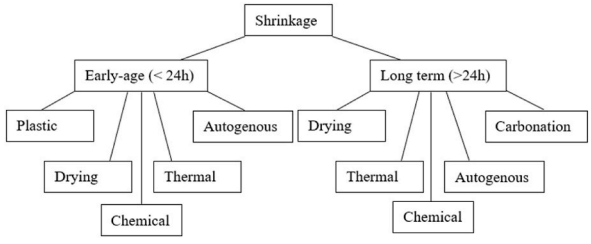


Fig. 1. Respective shrinkage mechanisms, adapted from Ref. [19].

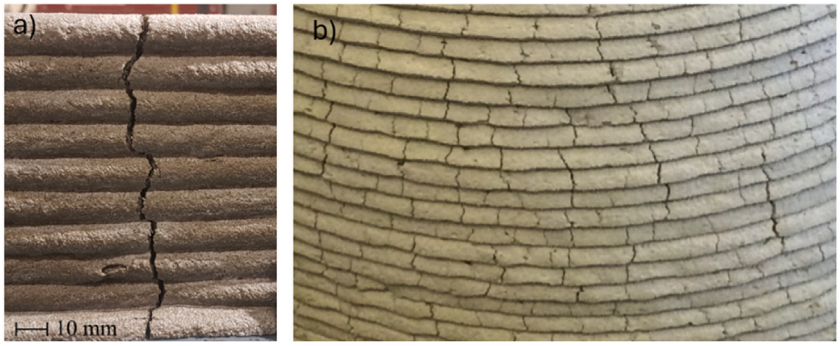


Fig. 2. a) PSC present in a laboratory specimen [50] and b) excessive early-age shrinkage in a 3D printed element provided without scale [63].

elements particularly vulnerable to PSC when shrinkage is restrained, as a low tensile strength is associated with fresh concrete [50]. Restrained shrinkage induces internal tensile stress that, if it exceeds the tensile capacity of concrete, will lead to cracking [20]. Fig. 2 shows a) a typical plastic shrinkage crack present in a laboratory specimen printed by Moelich et al. [50] and b) excessive PSC [63]. PSC is detrimental to the durability of 3DCP structures as it weakens the structure and allows for the ingress of aggressive agents that weaken the concrete and reinforcing steel if included [18,65]. Excessive early-age shrinkage and PSC may, to an extent, be mitigated by adapting the mix design, by including SAP polymers, or by applying effective curing techniques [63,66,67]. However, Moelich et al. [14] found that using short polypropylene microfibers was the only mitigation strategy that was 100 % successful in mitigating and PSC of printed members.

3.1.2. Drying shrinkage

Drying shrinkage is caused by the progressive loss of moisture as the concrete starts to gain stiffness [68]. It initiates at an early-age (<24 h) and persists long-term (>24 h) in some cases up to a few months after printing, depending on environmental exposure and material properties. Free water evaporates through the surface, particularly ay elevated temperature and low relative humidity, as the hydration reaction proceeds and a solid matrix is formed. The evaporation of free water at the surface caused early-age drying shrinkage. As the hydration reaction proceeds, absorbed water held in tiny capillaries by hydrostatic tension is also reduced as the drying of concrete continues [69]. This loss of free and absorbed water results in a volumetric reduction of concrete, known as long-term drying shrinkage [68,70,71].

The magnitude of drying shrinkage is closely linked to rate and extent of moisture evaporation [71] while the rate and extent of moisture evaporation are influenced by the pore structure of the printed material. Consequently, drying shrinkage is directly influenced by environmental conditions. 3DPC typically exhibits reduced w/c ratios that result in a finer capillary network. Higher capillary tension during drying is induced that results in increased drying and autogenous shrinkage. Additionally, during pumping and extrusion air bubbles accumulate near the surface as the nozzle moves, thereby promoting increased evaporation from the surface, which further increases drying shrinkage [71]. Similar to plastic shrinkage, drying shrinkage can also result in cracking of concrete elements if inadequate consideration is given to sufficient curing techniques [70].

3.1.3. Autogenous shrinkage

Autogenous shrinkage is defined by Jensen et al. [72] as: “The bulk deformation of a closed, isothermal, cementitious material not subjected to external forces.” and similar to drying shrinkage it bridges both early-age and long-term shrinkage. It occurs because of

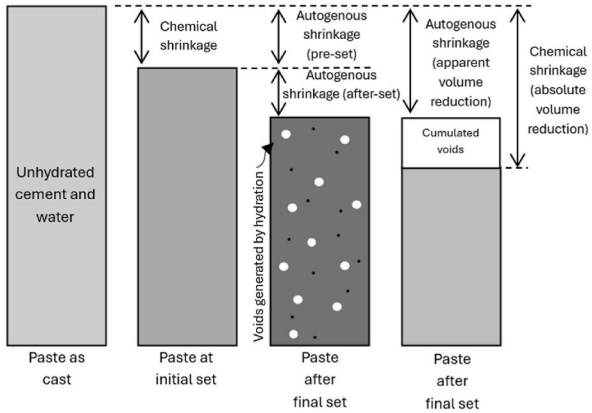


Fig. 3. Relationship between chemical and autogenous shrinkage [74].

water absorption as a result from the hydration reaction that results in internal structural changes. Autogenous shrinkage is the driven by chemical shrinkage and the process of self-desiccation [73]. Hence, chemical shrinkage and autogenous shrinkage are closely related, but not the same, which makes it difficult to distinguish between the two shrinkage mechanisms. However, Fig. 3 describes the relationship between autogenous and chemical shrinkage [74]. This figure shows that chemical shrinkage is the same magnitude as autogenous shrinkage up to the pre-set of concrete. After the final set, autogenous shrinkage becomes a fraction of chemical shrinkage, and the voids within the matrix after the final set account for the difference between the two mechanisms of shrinkage [73].

In various studies, autogenous shrinkage is considered negligible when compared to the magnitude of drying shrinkage [75]. However, in the context of the relatively low water-to-cement (w/c) ratio present in 3DCP mixes and ultra-high-performance concrete (UHPC) mixes, the magnitude of autogenous shrinkage may significantly contribute to the total shrinkage magnitude. It is, therefore, imperative to consider the influence of autogenous shrinkage of mixes with low w/c ratios, typically less than 0.3, as it increases the risk of cracking in restrained elements [72,76,77].

Water-reducing agents or superplasticisers are also commonly used in 3DCP to achieve and maintain the required rheological properties for a successful extrusion [78]. Varying impacts of superplasticizers on shrinkage have been reported, with studies reporting that they could reduce shrinkage depending on the age of the concrete. Cartuxo et al. [79] reported a 44 % increase in shrinkage at 7 days, while an insignificant 2 % reduction was noted at 28 days. The early-age increase in shrinkage is attributed to the altered distribution of water particles, which promotes evaporation and the subsequent early-age shrinkage. This is in accordance with the findings of Matias et al. [80] where they also reported a more profound influence of superplasticisers on early drying shrinkage than at a later stage. Hence, the effect of superplasticizers on the shrinkage of 3DPC also needs to be investigated, as there are uncertainties relating to their influence on shrinkage.

Four approaches have been identified to reduce the development of autogenous shrinkage in UHPC, namely: 1) regulating the hydration reaction, 2) controlling internal relative humidity (IRH), 3) providing internal restraint, and 4) introducing internal forces [81]. These approaches, aimed at reducing autogenous shrinkage for UHPC, can be used to mitigate autogenous shrinkage in 3DPC. Firstly, incorporating SCMs that exhibit reduced hydraulic reactivity and calcium content, compared to OPC, in printable mixes stalls the hydration reaction, which is the primary driver of autogenous shrinkage [82,83]. This contrasts with the findings of Al-Moudi et al. [26], where silica fumes increased the plastic shrinkage of conventional concrete. Secondly, self-desiccation occurs because of a reduction of IRH. Adding agents that promote internal curing by gradually releasing curing water contributes to maintaining the IRH. Internal curing agents include superabsorbent polymers (SAP) and porous fine aggregates [84]. Thirdly, internal restraint to autogenous shrinkage is provided by incorporating lightweight aggregate or reinforcing steel fibres that counteract shrinkage-induced stresses [41,85]. Lastly, internal forces can be introduced by adding shrinkage-reducing agents or introducing expansive agents that counterbalance autogenous shrinkage [86]. Although expansive agents reduce autogenous shrinkage, they may negatively influence the mechanical properties by altering the microstructure of the concrete [87].

3.1.4. Chemical shrinkage mechanisms

Zhang et al. [88] defined chemical shrinkage as the absolute internal volume change of cement paste resulting from the hydration of cementitious materials. In essence, the volume of the hydration products of the cement reaction is less than the volume of the reactants [89]. The volume reduction occurs immediately after adding water to the mix and remains prominent during the first hours and days [71]. Hence, it is both an early-age and a long-term shrinkage mechanism, as illustrated in Fig. 2. Although autogenous shrinkage is a fraction of chemical shrinkage after the final set [74], it is studied independently of chemical shrinkage mechanisms. The fundamental method of measurement also varies from that of autogenous shrinkage. Fully saturated cement paste is placed in a glass beaker, where the thickness of the paste should be thin enough to allow for water penetration throughout the paste. The chemical shrinkage is determined by measuring the change in water level, which indicates chemical shrinkage. This method is known as the dilatometry measurement method. Hence, chemical shrinkage is often not expressed as a strain (m/m or mm/m) but rather a specific volume per unit of mass, for instance, mm^3/g_{cement}. However, in a study by Khatib et al. [90] the volume per unit mass was converted to a magnitude of strain to allow comparison to other shrinkage mechanisms.

3.1.5. Thermal shrinkage and expansion

Thermal shrinkage, occurring shortly after extrusion as an early-age shrinkage mechanism, refers to the volumetric reduction resulting from temperature fluctuations attributed to the hydration reaction [43]. Initially, concrete experiences an increase in temperature when cementitious materials react with water, producing heat from this exothermic reaction [57]. The heat produced by the hydration reaction significantly influences early (<24 h) thermal behaviour, especially in mixes with low water to cement ratios as. Large concrete elements produce substantial amounts of heat [57]. This gives rise to differential thermal gradients and the consequent differential stresses between the core and surface zones of the element that could result in cracking. The subsequent cooling, once the peak heat of hydration temperature is reached, can induce high tensile stresses due to thermal contraction.

In the long-term (>24 h), thermal expansion and shrinkage result from heating and cooling due to daily and seasonal temperature cycles. While the thermal shrinkage related to the hydration reaction is considerable, later-stage thermal shrinkage or expansion related to ambient temperature fluctuations may induce significantly larger tensile or compressive stresses within the concrete. Temperature differences are usually avoided for printed elements in lab conditions where temperature remains constant. However, once the extrusion of elements takes place in an uncontrolled environment, temperature-related shrinkage and expansion pose a challenge to the risk of cracking.

3.1.6. Carbonation shrinkage

Cement paste in concrete typically contains free calcium hydroxide ($Ca(OH)_2$) that is a product of the hydration reaction [57,91]. In the presence of water, $Ca(OH)_2$ reacts with atmospheric carbon dioxide (CO_2), which results in a volumetric reduction of the cement paste known as carbonation shrinkage [56]. Research has shown that the atmospheric relative humidity, atmospheric CO2, paste porosity, moisture content, binder type and composition influence carbonation shrinkage [92,93]. However, carbonation shrinkage has a relatively small contribution to total shrinkage, especially since the ambient CO_2 cannot penetrate deep into high-quality and low-porosity concrete [94]. Surface zones are, therefore, mainly influenced by carbonation shrinkage as the carbonation depth varies from millimetres up to a few centimetres [57]. Lastly, since carbonation shrinkage is a surface phenomenon that is slow to progress, typically progressing over months to years, its classification as a long-term shrinkage mechanism is consistent as it has little early-age contribution to shrinkage. For more detailed elaboration of the mechanism of carbonation shrinkage and cracking the reader is referred to Ref. [56].

3.2. Reported shrinkage strains

While identical shrinkage mechanisms described in the preceding section occur in printed and cast concrete, their magnitudes differ due to the respective construction processes utilized. Also, the contribution of the respective mechanisms to the total magnitude shrinkage may vary between printed and cast concrete as there are fundamental differences in factors that influence shrinkage. Table 4 summarises the underlying differences in shrinkage mechanisms between printed and cast concrete.

The reported shrinkage values are summarized in Tables 5 and 6 for printed and cast concrete, respectively. In Fig. 4, p and c denote printed and cast concrete, respectively, and DS, PS and AS represent drying, plastic and autogenous shrinkage, respectively. Also, each bar's darker and lighter shade indicates the maximum and minimum shrinkage reported for each study. In Tables 5 and 6, T represents ambient temperature, RH represents the relative humidity, and V indicates the artificially induced wind speed. In relevant literature, the definition of zero time differs between studies where zero time typically represents the time interval to the first shrinkage measurement. For instance, Yoo et al. [98] defined zero time as the time interval from casting to the final setting time of concrete as this study aimed to exclude early-age shrinkage related to plastic shrinkage. In contrast, Moelich et al. [14] defined zero time as the time interval from the completion of the print up to the initial shrinkage measurement. Also, zero time of drying shrinkage measurement of cast concrete is typically set at demoulding which varies between 24 and 168 h from casting as specified in the respective standards. The zero times in Tables 5 and 6 are as reported in the respective reviewed study. It should be noted that in Tables 5 and 6, the absence of data indicates that a specific parameter was not reported in the corresponding reviewed study. Lastly, it should be noted that although drying shrinkage strains are measured and reported, a fraction of this may be attributed to autogenous shrinkage where ineffective curing occurs [99].

Fig. 4 summarises the reported shrinkage strains from Tables 5 and 6 Fig. 4 shows that 3DPC exhibits plastic and drying shrinkage strains significantly more than that observed for cast concrete, particularly when comparing plastic and drying shrinkage and their combined effects. The relevant literature reported an average drying shrinkage of approximately 1700 μm/m for 3DPC compared to 800 μm/m for cast concrete. As for plastic shrinkage, a 6500 μm/m strain is reported for 3DPC, compared to an average of 900 μm m/m for cast concrete. While this comparison may appear simplistic given the numerous factors that vary between each study, relevant literature consistently indicates that 3DPC is more susceptible to significant shrinkage than its cast counterpart. However, by comparing the autogenous shrinkage results from cast and printed concrete in Fig. 4, it is evident that cast UHPC (see Table 3) mixes exhibit significantly larger autogenous shrinkage than that reported for printed concrete.

3.3. Factors that influence shrinkage of 3DPC

In addition to the factors that influence shrinkage, included in Tables 5 and 6, other contributing factors are not included. All factors that influence the shrinkage of printed members are discussed in subsequent sections. While many of these factors also influence cast concrete, the factors that influence shrinkage differ slightly from 3DPC.

3.3.1. Environmental and curing conditions

Curing conditions directly impact the hardening of cement paste, which subsequently affects the development of the microscopic

Table 4
Differences in shrinkage mechanisms between printed and cast concrete.

Shrinkage mechanism	3DPC	Cast concrete
Plastic	Increased due to large surface exposure. Enhanced by differential drying and evaporation between layers [14].	Predominantly surface shrinkage for large exposed surfaces like slabs is commonly managed by effective curing [95].
Drying	Shrinkage occurs anisotropically relative to the direction of printing [96].	Shrinkage occurs relatively uniformly compared to 3DPC, although differential shrinkage occurs in cast concrete [97].
Autogenous	Requires consideration due to the low w/c ratio used in mixes to obtain printability requirements [32].	It is commonly less pronounced in traditional cast concrete, except for UHPC, which has low w/c ratios [96].
Thermal	It is less significant due to its thin, layered nature, which reduces internal heat buildup and allows for effective heat dissipation [43].	More distinct, especially in mass concrete elements [43].

Table 5
Reported shrinkage strains for 3DPC.

Study	Curing conditions	Specimen size (mm × mm × mm)	a/b[a]	w/c[b]	w/b[c]	Method, device, resolution (mm)	Zero time in hours (t_o)	Type of shrinkage	Test range (d)	Reported shrinkage (μm/m)
Le et al. [96]	i) 20 °C, 100 % ii) Wrapped iii) 20 °C, 60 %	75 × 75 × 229 (cast mortar specimen)	1.5	0.37	0.26	BS-EN 12617–4:2002, (−), 0.001.	24	Drying	184	i) 180 ii) 570 iii) 855
Ma et al. [100]	T: 20 ± 1 °C RH: 60 ± 2 %	i) 25 × 25 × 280 ii) 40 × 40 × 160 iii) 70 × 70 × 280	1.6	i) 0.3 ii) 0.4 iii) 0.5	–	Non-standard, Vertical length comparator, 0.001.	–	Drying	90	i) 900 ii) 1300 iii) 1400
Zhang et al. [101]	Cured for initial 7 days, thereafter: T: 20 ± 2 °C RH: 60 ± 5 %	60 × 60 × 300	1.0	0.35	0.35	GB/T50082-2019, HSG, 0.001.	168 (after curing)	Drying	60	Min 1300 Max 2400
Slavcheva [102]	T: 20 ± 2 °C RH: 20, 30, 50,70 %	10 × 40 × 160 (cast mortar specimen)	–	i) 0.4 ii) 0.34 iii) 0.24	–	Non-standard, comparator gauge, 0.01.	24 (after curing)	Drying	28	At 20 % RH: Min 1360 Max 2400
Duan et al. [24]	–	25 × 25 × 285 (cast mortar specimen)	1	0.35	0.35	ASTM C1148, Length comparator, (−).	–	Drying	28	Max 1400
Markin et al. [60]	T: 30 ± 2 °C RH: 40 ± 2 % V: 5 ± 0.3 m/s	100 × 75 × 500	1.82	0.51	0.46	Non-standard, DIC, (−).	1	Plastic	0.138	Min 870 Max 140
Han et al. [103]	T: 20 ± 1 °C RH: 60 ± 2 % V: 3.5 m/s	60 × 80 × 250	1	–	0.36	Non-standard, DIC, (−).	0.33	Plastic	0.10	Min 6000 Max 11000
Moelich et al. [14]	T: 25 ± 1 °C RH: 60 ± 5 % V: 3.6 ± 0.3 m/s	Multiple extruded lengths: L1 = 200 mm, 20 layers L2 = 300 mm, 20 layers	1.40	0.45	0.31	Non-standard, DIC, 0.01.	0.2	Plastic	0.10	Min 5000 Max 7000
Federowicz et al. [104]	T: 20 ± 3 °C RH: 60 ± 5 %	15 × 35 × 1350	–	0.34	0.24	Non-standard, Optical sensors, 0.002.	0.5	Plastic and drying combined	28	Max 4800
Van der Putten et al. [20]	T: 20 ± 3 °C RH: 60 ± 5 % T: 20 ± 3 °C RH: 65 ± 5 %	Single layer: l = 150 mm Corrugated tube, l = 420 ± 5 mm	2	0.37	–	Non-standard, DIC, (−). ASTM C1698, Digital dilatometers, (−)	0.167 0.167	Plastic and drying combined Autogenous	5 7	4000 1500
Rahul et al. [105]	Sealed in foil tape	40 × 40 × 160 (cast mortar specimen)	1.7	0.7	0.35	(−), DEMEC, (−)	24	Autogenous	31	Max 250 Min 100

[a] aggregate-to-binder.
[b] water-to-cement.
[c] water-to-binder.

pore structure of concrete. This, in turn, influences both mechanical and shrinkage properties development over time as variables such as temperature, humidity and airflow are kept constant [44]. The ability to achieve the desired properties of concrete is related to the length and starting point of curing. Curing is typically divided into internal and external curing, with both methods having equal importance in assuring shrinkage reduction of 3DCP elements [116].

Internal curing enhances hydration by providing additional moisture during the drying phase. Research has shown that it decreases the rate of internal relative humidity (IRH) reduction, which in turn mitigates the autogenous and thermal shrinkage [117,118]. To enhance internal curing, researchers have incorporated SAP in concrete mixes to assist in retaining internal curing water, thereby avoiding the self-desiccation of concrete [119]. Light-weight aggregates are also incorporated into concrete mixes to improve the internal curing ability of a particular concrete mix [41]. The effectiveness of an internal curing regime is related to the replacement

Table 6
Reported shrinkage strains for conventional concrete mixes.

Study	Type of concrete	Curing regime	Sample dimensions (mm × mm × mm)	a/b	w/c	w/b	Method, device, resolution (mm)	Zero time (t_0)	Type of shrinkage	Test range (d)	Reported shrinkage (mm/m)
Sagoe-Crentsil et al. [106]	Recycled aggregate	T: 23 °C RH: >95 %	75 × 75 × 280	–	0.76	–	AS 1012, (−), 0.002.	168	Drying	365	700–1000
	Conventional				0.7–0.74	–		168	Drying	365	600
Collins et al. [107]	Alkali activated slag	T: 23 °C RH: 50 %	Ø63 × L98	5.4	–	0.5	AS 1012, (−), 0.002.	168	Drying	365	2000
	Conventional			5.4	0.5	–		168	Drying	365	500
He et al. [108]	Lithium slag concrete	T: 20 ± 21 °C	100 × 100 × 300	3.9	0.35	0	(−), Dial indicator, (−).	168	Drying	150	250
	Conventional	RH: 60 ± 5 %		3.9	0.5						
						0.35		168	Drying	150	300
Meng et al. [83]	Class C fly ash binder	T: 20 °C RH: 50 %	100 × 100 × 250	–	0.2	–	ASTM C596, (−), (−).	168	Drying	98	500
	Conventional			–	0.2	–		168	Drying	98	600
Younis et al. [109]	Conventional	T: 20 ± 1 °C RH: ±50 %	100 × 100 × 500	4.22	1.04	0.36	ASTM C157/C157M, (−), (−).	24 24	Drying Drying	56	650
	Seawater concrete									56	620
Turcry et al. [110]	Self-consolidating concrete	i) T: 20 ± 1 °C RH: 50 ± 2 % ii) T: 20 ± 1 °C RH: 50 ± 2 % V = 5 m/s	70 × 70 × 280	3.3	0.57	0.41	Plastic shrinkage box, Lasor sensors, (−).	0.33 0.33	Plastic Plastic	1 1	100 1200
Gonzalez-Coromina et al. [111]	Recycled aggregate	T: 20 ± 0.5 °C RH: 50 ± 2 %	150 × 150 × 600	5.3	0.29	–	(−), Shrinkage box with LVDTs, (−).	0	Plastic	0.5	400–600
	Conventional			5.3	0.29	–		0	Plastic	0.5	200
Etxeberria et al. [112]	Slag and seawater replacement of binder and water	T: 20 °C RH: 60 %	150 × 150 × 600	6.5	–	0.45	Identical so Saliba et al. [42].	0.5	Plastic	0.29 (7 h)	380
Mazloom et al. [113]	Silica fume binder	T: 20 ± 2 °C RH: 50 ± 5 %	Ø80 × L270	3.7	0.43	0.35	(−), DEMEC[a] gauge, 0.01.	168	Autogenous	600	200
	Conventional			3.7	0.35	–		168	Autogenous	600	300
Yoo et al. [98]	UHPC with SRA	T: 20 ± 1 °C and sealed in polyester film	100 × 100 × 1000	1.1	0.25	-	(−), Dumbbell-shaped strain gauge, (−).	Final setting time	Autogenous	30	530
	UHPC										780
									Autogenous	30	
Ma et al. [114]	UHPC	–	100 × 100 × 700	1.5	0.268	0.206	(−), Shrinkage box with LVDTs, (−).	–	Autogenous	20	400
Yalçınkaya, Yazıcı [115]	UHPC	i) T: 20 °C, ii) T: 30 °C, iii) T: 40 °C, fully sealed specimen	25 × 25 × 290	1.26	0.25	0.2	ASTM C169, Lasor sensors, (−)	IST[b]: i) 4.32 ii) 2.5 iii) 2.25	Autogenous	1 day	i) 400 ii) 600 iii) 800
Khatib et al. [90]	Mortar	closed environment	250 ml glass bottle	–	0.56	0.45	ASTM C1608, (−).	0	Chemical	90	1700

(*continued on next page*)

Table 6 (*continued*)

Study	Type of concrete	Curing regime	Sample dimensions (mm × mm × mm)	a/b	w/c	w/b	Method, device, resolution (mm)	Zero time (t_0)	Type of shrinkage	Test range (d)	Reported shrinkage (mm/m)
	Mortar		25 × 25 × 300	–	0.56	0.45	ASTM C192, DEMEC, 0.001.	24	Autogenous	90	1500
Saliba et al. [42]	Conventional	T: 20 ± 1 °C RH: 50 ± 5 %	70 × 70 × 280	3	0.43	0.3	(−), Plastic shrinkage box with laser sensors, (−).	0.5	Plastic	2	1500
		fully sealed specimen	70 × 70 × 280				(−), Plastic shrinkage box with LVDT, (−).	24	Autogenous	180	180
		T: 20 ± 1 °C RH: 50 ± 5 %	70 × 70 × 280				(−), LVDTs, (−).	24	Drying	180	220

[a] Demountable Mechanical Stain Gauge.
[b] Initial Setting time.

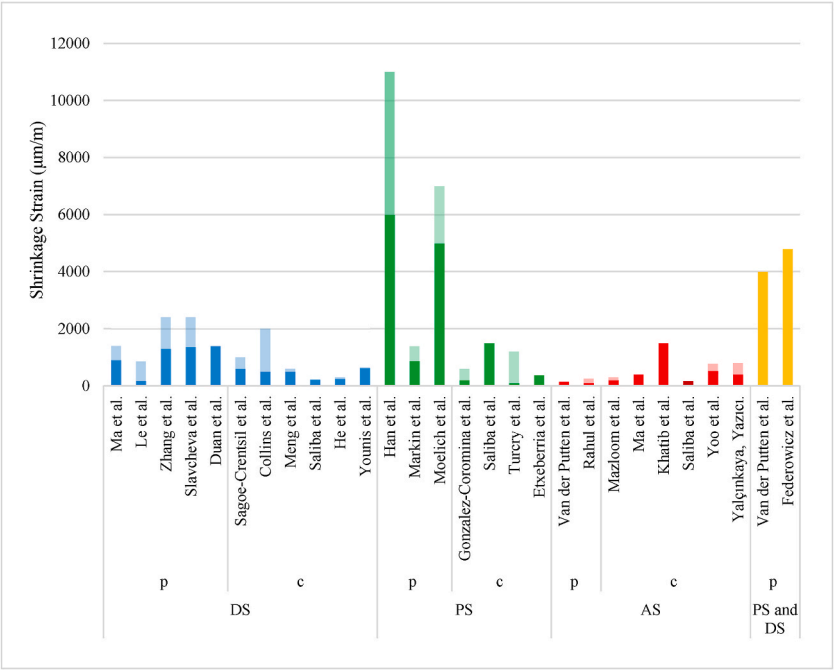


Fig. 4. Summary of reported shrinkage strains for printed and cast concrete.

ratio and particle size of SAP in high-performance concrete, with increased particle size and higher replacement ratios resulting in reduced autogenous shrinkage.

External curing entails applying external processes to ensure sufficient hydration of concrete during the drying phase. Ambient conditions determine the effectiveness of external curing and should be accounted for to mitigate plastic, autogenous and drying shrinkage [120]. Shahmirzadi et al. [116] investigated the influence of RH and temperature on autogenous shrinkage, moisture loss and the consequent drying shrinkage of cementitious materials. This study found that increased ambient temperatures and a reduced RH percentage exacerbate drying shrinkage. Specifically, curing at 45 % RH and 35 °C resulted in greater shrinkage when compared to curing at 50 % RH and 24 °C, highlighting the sensitivity of 3DPC to temperature and RH variations. Additionally, Moelich et al. [50] developed a crack-risk prediction model for 3DPC exposed to different environmental conditions. This study found that higher evaporation rates, because of increased curing temperature and lower RH levels, directly increase the risk of PSC. Additionally, researchers introduced artificial air flow with the use of fans or wind tunnels to simulate more extreme environmental conditions that induce higher evaporation rates [14,103,121]. As 3DPC typically exhibits a low bleeding rate in comparison to cast concrete, limited protection of pore water evaporation is provided. Consequently, surface water evaporates rapidly which results in the buildup of capillary pressure that pulls the particles together. This in turn results in increased plastic shrinkage and an increased risk of PSC [50].

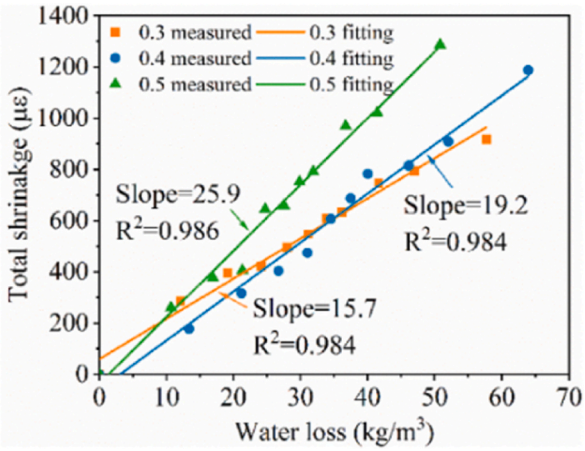


Fig. 5. Correlation between the total water loss and total shrinkage [100].

The magnitude of shrinkage is fundamentally related to the amount of water lost as shown in Fig. 6 [100]. It should be noted that the amount of water consumed, due to evaporation and hydration, provides a more accurate relationship with shrinkage as shown in Fig. 5 than the relationship between only evaporated water and shrinkage.

Drying and plastic shrinkage are both influenced by environmental conditions, as these mechanisms are driven by the loss of water through the surface of concrete. Therefore, the extent of plastic and drying shrinkage is intrinsically linked to the significant area of the exposed surface in relation to specimen volume, i.e., the surface-to-volume (s/v) ratio [122]. Yuan et al. [68] investigated the influence of mix properties and found that an increase in sand-to-binder ratio reduced shrinkage. An approximate 20 % reduction in drying shrinkage is measured up to 90 days with an increase of s/b ratio from 0.8 to 1.2.

3.3.2. Mix proportions

As previously mentioned, printable mix proportions vary significantly from cast concrete. This variation in mix design is to ensure that rheological properties align with appropriate fresh state requirements for pumpability, buildability and extrudability [123,124]. Printable mixes are characterised by a low dynamic yield stress to assist with sufficient pumping, but high thixotropic behaviour is also present to support the weight applied by subsequently extruded filaments [125,126]. 3DCP mixes typically contain more binder and fine aggregate than cast concrete to achieve the sought-after fresh-state properties. This is evident when comparing the aggregate to binder ratios reported in Tables 5 and 6

As summarized in Table 5, most printable mixes exhibit relatively low w/c ratios ranging from 0.3 to 0.4. Research conducted on cast concrete found that mixes with low w/c are prone to significant autogenous shrinkage [16,71,116]. Although autogenous shrinkage of printed concrete has been investigated to a lesser extent, printable mixes may exhibit significant autogenous shrinkage due to 3DPCs relatively low w/c ratios. Hence, the w/c and w/b ratios are of importance as they not only influence shrinkage but also affect mechanical and workability properties [127].

Typical 3DCP mixes have an aggregate volume percentage of approximately 50 %, while the total aggregate volume of cast concrete accounts for approximately 70 % or more [128]. Printable mixes also incorporated higher quantities of fine aggregates, contributing to greater water demand due to their larger surface-to-mass ratio. The high quantity of fine aggregates, in combination with the incorporation of SCMs like silica fumes, have been found to further increase the risk of PSC due to the smaller pore space associated with higher capillary pressure in the fresh concrete [27,129]. Duan et al. [24] investigated the influence of incorporating metakaolin (MK), another SCM, as a rheology modifier on the fresh and hardened state properties. Although MK was used as a rheology modifier, they found that its inclusion resulted in significant drying shrinkage reductions at a specimen age of 30 days. Bhattacherjee et al. [130] also investigated the influence of mix design variation on drying shrinkage and found that the final magnitude of drying shrinkage strain decreased from 1000 μm/m to 800 μm/m with an increased aggregate-to-binder ratio from 1.5 to 3 for printable mixes.

In the context of 3DCP, Rahul et al. [105] found that the incorporation of recycled coarse aggregates (RCA) had a favourable effect as it significantly reduced autogenous shrinkage. This study measured the autogenous shrinkage of a cast printable mix. Cast specimens neglect the influence of printing-related parameters on autogenous shrinkage. Nevertheless, incorporating RCAs improved internal curing, which is attributable to the higher water absorption properties of RCAs compared to natural coarse aggregates.

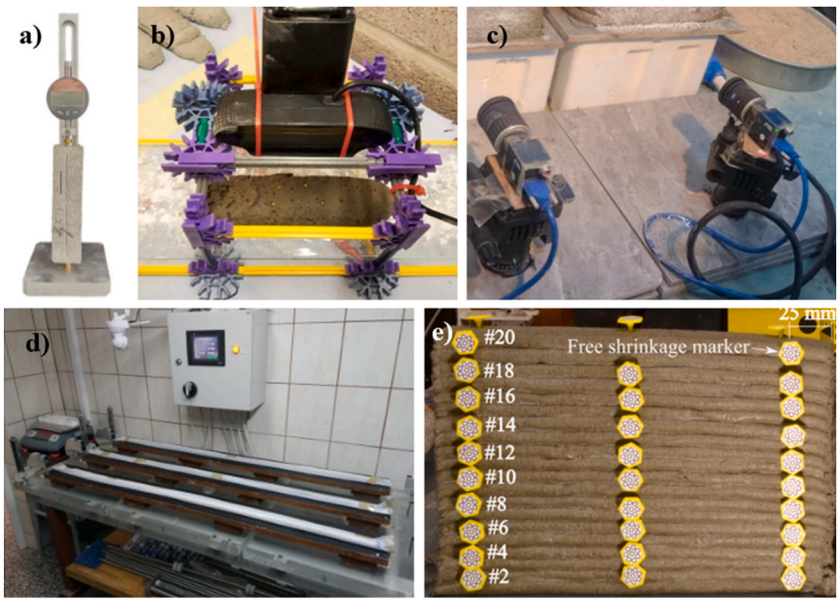


Fig. 6. Shrinkage measuring methods used in reviewed literature: a) Vertical length comparator by Ma et al. [100], b) DIC by Van der Putten et al. [20], c) DIC by Han et al. [103], d) Optical sensors used by Federowicz et al. [104] and e) shrinkage markers used for DIC by Moelich et al. [14].

3.3.3. Method of testing and zero time

Currently, there is no standard method for characterising the shrinkage of 3DPC. Consequently, researchers implemented a variety of shrinkage measuring methods, with different methodologies as outlined in Table 5. Some researchers followed the testing procedures specified for conventional concrete, whereas Le et al. [96] used BS-EN 12617–4:2002 [131] and Zhang et al. [101] used GB/T50082-2019 [132] while other researchers used non-standard methods. Sample preparation also varies between studies as drying shrinkage is measured from cast mortar specimens in some studies [24,96,102], while others opted to measure the shrinkage of printed specimens. Using cast specimens will provide an approximate shrinkage strain but is not an exact representation of 3DPC as the extrusion process and environmental conditions directly post-printing are neglected [101]. Plastic shrinkage is commonly measured with the use of digital image correlation (DIC) or distributed optical fiber sensors (DOFS) [133]. DIC monitors the movement of markers through successive images taken at fixed time intervals after deposition [14,20,60,103]. DIC is favoured above traditional measuring equipment like linear variable differential transformers (LVDTs) due to its non-intrusive nature. It does, however, require extensive software and data analysis and expensive photographic image processing equipment [101]. DIC should also be conducted in stable light and vibration-free environments [133]. DOFS have widely been used to measure long term shrinkage strains, corrosion depth and temperature changes of hardened concrete [134]. As the specimen experiences shrinkage, the embedded DOFS undergo strain. Inadequate bonding of optical fibers could result in strain loss or errors. The wavelength change of the light passing in the DOFS is used to calculate strains in different positions along the optical fibers [133]. Similar to DIC, DOFS requires the use of expensive equipment and complex data analysis. Also, these optical fibers could easily be broken or damaged during embedding or curing. LVDTs present a challenge in mounting printed specimens in the fresh state and are consequently preferred for drying shrinkage measurements over extended periods [31,135]. High-accuracy measurements are obtainable with the used of LVDTs, DIC and DOFS. Other researchers used lower resolution measurement equipment like a DEMEC strain gauge or a vertical length comparator to measure shrinkage where these shrinkage measuring equipment is typically used for standardised concrete specimen sizes. However, the use of a DEMEC gauge requires manual measurements and is not preferred due to the potential variability of manual measurement.

The respective test methods used to quantify the shrinkage deformation in the reviewed literature are shown in Fig. 6. Given the range of testing methods used to measure the shrinkage of 3DCP elements, discrepancies may easily occur, leading to variations in the reported shrinkage strains. Directly after deposition, while concrete is still in a plastic state and due to the thixotropic nature of 3DCP, any handling-related agitations could result in deformations and even collapse of printed elements. This underscores the importance of careful handling and sample preparation of specimens to ensure an undisturbed printed structure to achieve accurate shrinkage measurements [136]. Furthermore, it is proposed that researchers avoid placing markers on the interlayer to protect the interlayer properties thereof [137]. Markers should be made of non-reactive materials with the concrete to ensure that the markers do not alter concrete properties.

As seen in Tables 5 and 6, the studies define different zero times, leading to further inconsistencies between reported shrinkage stains. Since shrinkage is a continuous phenomenon starting directly after the deposition of the filament, the definition of zero time will significantly influence the captured shrinkage. In the reviewed literature, the definition of zero-time varies between studies as researchers aim to isolate a certain mechanism of shrinkage. For instance, Zhang et al. [101] defined zero time at 7 days after curing thereby allowing for most part of chemical and autogenous shrinkage to have ocurred. This allows for the measurement of mainly the long-term drying shrinkage. However, when considering plastic shrinkage, a shrinkage mechanism that is highest about 2 h after printing, it is essential to define the shortest possible zero time [14,40]. This ensures complete characterisation of plastic shrinkage up to the final setting time of concrete of the specific mix. It is, however, challenging to define shrinkage zero time shorter than 10 min as concrete needs to be mixed and extruded before shrinkage measuring can commence. Therefore, the definition of zero-time needs to be carefully considered to ensure that the wanted shrinkage mechanism is characterized.

3.3.4. Admixtures, shrinkage-reducing agents, expansive agents and fibers

Admixtures are commonly incorporated into printable concrete mixes to achieve sufficient rheological and fresh-state properties. Zhu et al. [44] identified the commonly used admixtures to reduce shrinkage as expansion agents [138], shrinkage reduction admixtures (SRAs) [139] and water-reducing agents [140].

SRAs are most effective at reducing shrinkage at an early age, as the hydration reaction is delayed [141]. The hydration reaction consumes water, which results in a decrease in IRH. As previously explained, the reduction in IRH leads to the development of capillary pressures within the concrete pores, which is considered one of the main mechanisms that result in shrinkage. Therefore, by delaying the hydration reaction, a high IRH is maintained, and shrinkage is mitigated [142,143]. Federowicz et al. [104] investigated the influence of including 2 % and 4 % SRA by cement mass, comparing this to the shrinkage of a sample cured under a protective layer of foil. Their findings highlighted a shrinkage reduction of 23 % in the samples that incorporated 4 % SRAs, while a smaller reduction of 7 % was present in samples with a 2 % SRA content. The foil-protected layer exhibited a significantly lower shrinkage rate and smaller magnitude than the uncured mixtures containing SRAs. Since the foil layer controls water loss to the environment, it is suspected that this shrinkage strain likely indicates the autogenous shrinkage of the concrete paste. This study concluded that curing had a more profound influence on the total magnitude shrinkage of 3DCP, reducing shrinkage from 5178 μm/m to 1031 μm/m. In addition, the use of SRAs has been reported by some researchers to slightly reduce the compressive strength and elastic modulus of tested samples at higher dosages [42], while others, such as Moelich et al. [40], found no significant difference in compressive strength properties at the use of smaller dosages. Given the high cost of SRAs and the limited effectiveness in mitigating shrinkage in comparison to other shrinkage mitigation techniques, like effective curing regimes, the viability of the use of SRAs in 3DCP warrants further investigation.

Due to the high cost of SRAs, researchers reduce the usage by compounding it with expansion agents [43]. The reaction between the cement and the hydration products result in a volume increase that counteracts shrinkage deformations, thereby reducing the risk of

shrinkage cracking [144]. However expansive agents work most effectively when the concrete is wet cured, otherwise enough crystals might not form to counteract shrinkage [145]. Another disadvantage of expansive agents is that at w/c ratios lower than 0.4, as typically prevalent in 3DPC mixes, the ability to mitigate shrinkage decreases. Consequently, Aghaee et al. [35] proposed the use of 10 % expansive agents with 1 % SRA or 0.125 % SAP to achieve a high thixotropic cement paste that exhibits favourable pumpability and buildability properties with low autogenous shrinkage and high strength.

3.3.5. Specimen size effect on shrinkage

The evaporation of moisture through the surface is intricately linked to the area of the exposed surface of 3DPC. Due to the absence of formwork, large surface areas are exposed to environmental conditions compared to volume. Elevated surface-to-volume ratios are correlated with increased water loss through the exposed surfaces, which results in increased drying shrinkage strains [44,100,146]. Shahmirzadi et al. [116] found that specimens with a lower surface-to-volume ratio tend to have better water retention capabilities, thereby reducing the total magnitude of shrinkage. This is in accordance with the findings of Samouh et al. [147] where they found that the magnitude of drying shrinkage of cylinders with a s/v ratio of 0.049 is lower than that of those with a s/v ratio of 0.102. Although the specimen size does not predominantly influence the total magnitude of drying shrinkage compared to environmental conditions, it still influences the shrinkage rate. This is evident as Bissonnette et al. [120] reported that a $4 \times 8 \times 32$ mm^3 specimen experienced a more significant shrinkage rate when compared to a $50 \times 50 \times 500$ mm^3 specimen. However, an equivalent final magnitude of shrinkage was reported for these two samples.

Analogous to drying shrinkage, the rate and magnitude of plastic shrinkage are similarly influenced by the s/v ratio of the specimen [148]. Also, considering 3DPC in the plastic state, the underlying layers carry the weight of the subsequently extruded filaments, where the weight of the overlying layers results in residual friction at the bottom layer. This residual friction is a source of constraint that could lead to cracking [103]. Han et al. [103] found that the bleeding rate and surface moisture significantly increased under the increased weight of upper layers, leaving the fresh concrete particularly vulnerable to plastic shrinkage. Moreover, in this study, an increase in the number of upper layers resulted in larger internal tensile stresses because of the development of capillary pressure. Increased principal tensile stresses are associated with an increased self-weight of upper layers, increasing the risk of PSC. This study concluded that the friction at the base during the printing and curing process needs to be minimized to reduce the risk of PSC.

3.4. Shrinkage prediction models

To date, a variety of shrinkage prediction models have been developed for cast concrete, mortar mixes and cementitious materials [149–151]. Similarly, Zhang et al. [152] developed a micromechanical model for autogenous and drying shrinkage of cast concrete. Mao et al. [153] developed autogenous and drying shrinkage prediction models for recycled aggregate concretes (RAC). These two models, for drying and autogenous shrinkage, respectively, are derived from similar frameworks developed by Zhang et al. [154,155], Guo et al. [156] and Fathifazl et al. [157]. However, in contrast to the three models used to develop the shrinkage model, the newly developed models are based on the two-phase shrinkage of recycled aggregates, where the shrinkage of natural coarse aggregates (NCA) and that of the residual mortar are calculated separately to determine the total shrinkage of RAC. These researchers sought to create an extensive shrinkage model by incorporating influential parameters, as discussed in Sections 3.3.1 to 3.3.5. Since only a few parameters needed to be regressed with previous test results, a more accurate model across a broader range of parameters is obtained. Detailed model derivations can be obtained from Mao et al. [153] Although models developed for cast concrete may assist in developing similar models for 3DPC, the underlying mechanisms of shrinkage vary along with key process-related parameters. By developing shrinkage prediction models for 3DCP like those developed for cast concrete, excessive shrinkage can be foreseen, and appropriate preventative measures can be implemented.

Ma et al. [100] developed a long-term shrinkage prediction model for 3DPC by refining and adjusting existing prediction models developed for cast concrete. Although the shrinkage mechanisms of 3DCP and cast concrete are similar, the printing process of 3DPC poses more specific characteristics that must be considered during model development. The model formulation is based on shrinkage mechanisms at 60 % RH, where shrinkage is primarily driven by capillary pressure (P_{cap}). It forms part of other important model parameters like the degree of saturation and bulk modulus. These parameters are used in a Bishop-Biot model, which is based on the concept of effective stress to address shrinkage associated with capillary pressure. Since effective stress considers fluid pressure within materials and the external loads applied, it can be used to accurately describe the deformation of saturated porous materials like concrete [158]. The developed shrinkage prediction model is shown in Equation (1):

$$\epsilon_s = \frac{P_{cap}.S}{3}\left(\frac{1}{K_b} - \frac{1}{K_s}\right) \tag{1}$$

where ε_s is the shrinkage strain, (P_{cap}) is the capillary pressure in GPa, S is the degree of saturation, K_b is the bulk modulus of the porous material and K_s is the solid body's bulk modulus typically ranges from 38 to 55 GPa. This variance is attributed to the bulk modulus of all solid phases in the system, while the main uncertainty pertains to the bulk modulus of the C − S− H compounds primarily due to its porosity [158]. Capillary pressure in Equation (1) is described using the Kevin Laplace equation [159]:

$$P_{cap} = \frac{In(RH).R.T.\rho_w}{M_w} \tag{2}$$

Where $In(RH)$ is the experimentally determined IRH; R is the universal gas constant, 8.314 J/(mol. K); T is the absolute temperature; ρ_w is the density of water, 1000 kg/m^3; M_w is the molar mass of water, 0.01802 kg/mol

The degree of saturation is determined by one of three methods: the mass change method, the Powers' model [160] and the degree of saturation based on the weight of a specimen under varying conditions [161]. These models mainly consider the influence of evaporation on the degree of saturation, while the hydration effect is less pronounced in the model developed by Ma et al. [100], since 3DPC is exposed to drying conditions directly after printing, the effects of evaporation and hydration need to be considered. Consequently, they calculated the degree of saturation as follows:

$$S = \frac{w_{liq}}{\rho_w} \Big/ \varphi \tag{3}$$

Where w_{liq} denotes the free water content, calculated as the difference between initial water content and the sum of hydration and evaporation-induced water losses, φ denotes the experimentally determined porosity.

This model made reasonable shrinkage predictions, with predicted shrinkage remaining within 20 % of the measured shrinkage, as shown in Fig. 7. However, the proposed model tends to underpredict occurring after 10 days from the initial measurement. These discrepancies relate to the model assumptions and simplifications made during model development.

Furthermore, Ghourchian et al. [48] used a generalised consolidation equation to model the development of capillary pressure in 3DPC elements as shown in Equation (4). This is particularly relevant to modelling shrinkage in printed elements as the development of capillary pressure drives the plastic shrinkage of extruded specimens. The poromechanical model, typically used to describe the consolidation and drying of porous materials, is described as follows:

$$\left(\frac{\delta S_w}{\delta \psi_p} n_{cap} + \frac{S_w}{B}\right) \frac{\delta \psi_p}{\delta t} = \nabla \left[\frac{\vec{k}_{sat}}{\rho_f g} \left(\nabla \psi_p + \rho_f \vec{g}\right)\right] - \dot{m}_{hydr} \left(\frac{1}{\rho_f} - \frac{S_w}{\rho_s}\right) \tag{4}$$

Where S_w denotes the degree of saturation (−), $\dot{m}_{hydr}$ the rate of water consumption during hydration (kg/m^3.s), ρ_f and ρ_s the densities of solid and pore fluid, respectively (kg/m^3), k_{sat} the saturated permeability (m/s), n_{cap} the capillary porosity (m^3/m^3), ψ_p the pore fluid's pressure potential (Pa), B the bulk modulus (Pa) and g the gravitational constant (m/s^2). The term on the right defines the chemical shrinkage rate and is replaced with the chemical shrinkage term as stated in Powers' model as shown in Equation (5):

$$\dot{m}_{hydr} \left(\frac{1}{\rho_f} - \frac{S_w}{\rho_s}\right) = 0.2 \left(\frac{M_{cem}}{\rho_{cem}}\right) \frac{\delta \alpha}{\delta t} \tag{5}$$

Where M_{cem} denotes the initial mass of cement in the mixture (kg/m^3), ρ_{cem} the cement density (kg/m^3) and α is the degree of hydration (−).

The model specified in Equation (4) was implemented in a 2D and 3D finite element model within a discrete space and timeframe. It demonstrated high accuracy in predicting capillary pressure and evaporated water development of 3DPC specimens over time. Given that capillary pressure is a function of the physical properties of the fresh concrete, the environmental conditions and the print geometries, these parameters can be combined in future research to develop a plastic shrinkage prediction model for 3DCP [162] similar to that developed by Ghourchian et al. [163].

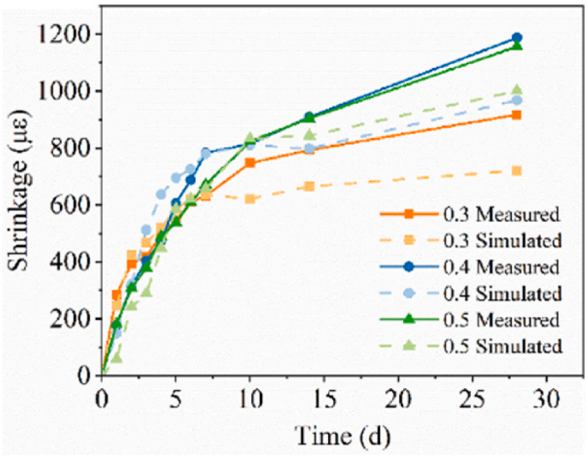


Fig. 7. Measured and simulated shrinkage strains at varying w/c ratios obtained from Ma et al. [100].

4. Shrinkage and cracking

Understanding the interplay between shrinkage and cracking is crucial in concrete applications, whether it be cast or 3DPC [20]. The degree of restraint, the magnitude of shrinkage and the stiffness of the concrete are all pivotal factors in the development of shrinkage-induced stresses. If these stresses exceed the tensile capacity of concrete, cracking will occur. It is vital to avoid cracking to prevent the intrusion of unwanted gases and fluids, which may lead to the corrosion of reinforcing steel and raise long-term durability concerns [20].

4.1. Unrestrained or free shrinkage

When shrinkage occurs unimpeded, a free or unrestrained shrinkage condition is achieved. In such cases, shrinkage does not result in cracking, as the buildup of internal stresses is avoided. However, achieving an entirely unrestrained shrinkage condition is practically challenging, especially in the fresh state of extruded concrete. Even when a virtually frictionless surface is created below the first layer, the weight of subsequently extruded surfaces creates a residual degree of restraint by the underlying layers.

The literature reports innovative solutions to create near frictionless surfaces for measuring the unrestrained plastic shrinkage of printed specimens. For instance, Markin et al. [60] created a low friction interface between the base and the first filament layer by covering the base with polyethene foil. This foil layer is then covered with oil, and another foil layer is placed on top of the oiled foil layer. In this study, a sample size of 100 mm × 75 mm × 500 mm was chosen to measure the unrestrained vertical settlement and horizontal shrinkage. The edges of the specimen exhibited larger vertical settlement compared to the centre of the specimen as a result of the quicker drying at the ends of the sample. As expected, the upper layers exhibited higher horizontal shrinkage than the bottom layers. This is attributed to the higher exposed surface area compared to the bottom layers.

Moelich et al. [14] also measured the unrestrained plastic shrinkage of individual 3DCP layers where layers are extruded on a base coated evenly with mould release oil. This study proposed covering the base with wax paper or plastic sheeting in future shrinkage tests to reduce shrinkage at this interface further, improving the accuracy of the unrestrained plastic shrinkage test. Free shrinkage was measured in the direction of printing for specimens with a length of 200 mm and 300 mm and varying heights. They found that a substantial amount of free shrinkage occurs within the first 2 h due to a moderate evaporation rate. Also, the rate of strain gain and peak strain is significantly larger than that of cast concrete.

In contrast to early-age shrinkage of 3DPC, free shrinkage conditions make it easier for hardened specimens with strength and stiffness to be sawn and handled. For instance, standards like BS-EN 12617–4:2002 [131], GB/T50082-2019 [132], BS EN 12390:16–2019 [164] and the ASTM C157 [165] have been used in literature to measure the free-drying shrinkage of concrete elements.

4.2. Restrained shrinkage and shrinkage cracking

Concrete's inherently low tensile strength, especially at early ages, renders it vulnerable to cracking under restrained shrinkage [69]. This vulnerability is mitigated to some extent by the relatively low stiffness of freshly extruded concrete. Specifically, the low stiffness at an early age diminishes shrinkage-related stress development. For instance, a given shrinkage strain may induce significantly larger stresses post-hardening than a similar strain during the pre-setting phase. This is attributed to the development of the elastic modulus once the concrete starts to harden. In addition, the corresponding tensile relaxation, which is related to its creep behaviour, contributes to alleviating internal shrinkage-induced stresses, whereby tensile creep may assist in reducing internal damage within restrained elements [84].

Cracks are typically a result of a combination of shrinkage mechanisms, including plastic, drying, autogenous and thermal, that do not act simultaneously but overlap in time to induce tensile stresses within the restrained printed elements [166]. The magnitude of these tensile stresses is directly proportional to the difference between free shrinkage (ε_f) and measured shrinkage (ε_a) as shown in Equation (6), and not the measured shrinkage strain [167].

$$\varepsilon_r = \varepsilon_f - \varepsilon_a \tag{6}$$

where ϵ_r is the restrained shrinkage. Restrained shrinkage can be quantitatively expressed with a restraint factor (R) as a fraction of free shrinkage, as seen in Equations (7) and (8):

$$\varepsilon_r = R\varepsilon_f \tag{7}$$

or

$$\varepsilon_a = (1 - R)\varepsilon_f \tag{8}$$

where R = 0 for free shrinkage and R = 1 for restrained shrinkage.

Importantly, shrinkage is restrained either by internal or external restraints [168], where internal restraints are caused by internal factors, like steel reinforcement, thermal or hygral gradients, inclusion of fibres, coarse aggregates or differential shrinkage. Conversely, external restraints arise from the external environment, like extrusion on older, stiffer concrete layers, extrusion on a near rigid surface, connection to other structural elements or print-geometry-related restraints.

For instance, shrinkage of 3DPC elements is frequently restrained by external restraints like the print geometry, base friction or

steel reinforcement [14]. Hoffman et al. [169] induced restrained shrinkage that ultimately resulted in PSC by providing external restraints when filaments were extruded on a lintel, an element considered nearly rigid relative to freshly extruded filaments, as seen in Fig. 8a. Due to this restraint, cracking was observed a day after printing as the mechanical properties have not yet developed. Small cracks were initially present in the wall above the lintel. These cracks developed into structural cracks over time and were a result of restrained plastic and autogenous shrinkage. Hoffman et al. stated that cracking occurred as a combination of 1) the presence of a rigid beam that induces restrained shrinkage, 2) the forces of friction between the concrete mix and the lintel beam surface also acted as shrinkage material and 3) since the left and right hand side of the wall section was printed in a single operation, their shrinkage involved uniform deformation to the centre of the left and right wall section respectively.

Zhang et al. [101] studied the influence of printing paths on the drying shrinkage of 3DCP mortar. Identical sample sizes of 60 mm × 60 mm × 300 mm were used in this test with a shrinkage measuring gage length of 250 mm. Notably, the transverse and oblique print paths, as illustrated in Fig. 8c, demonstrated shrinkage strains of 18.4 % and 11.8 % lower, respectively, than those of the longitudinal printing paths. This is suspected to be due to the expansion of micro cracks resulting from the printing process, which leads in the loss of total drying shrinkage strains.

Moelich et al. [14,64] explored different combinations of print geometry, base friction and reinforcement to induce PSC. In the first study [14] this entailed inserting steel rods into the concrete to induce restrained shrinkage as seen in Fig. 8b. The aim of this study was to investigate the influence of restrained shrinkage on the interlayer bond of printed concrete. In this study they found that shrinkage strain increases linearly from the one layer to the next as the distance (source of restraint) from the base increases. However, these strains are not transferred uniformly between interlayers and results in a discontinuity at the interlayer. These discontinuities attributed to slip at the interlayer. In another study by Moelich et al. [14] different combinations, shown in Fig. 9, are used to restrain shrinkage by either a) the printed geometry, b) a roughened base, c) inserting two steel rods connected to a rigid element, d) inserting four steel rods connected to a rigid element or e) inserting two steel rods and a notch midway between restraining rods [14]. Upon testing the respective restraint combinations, cracking was observed in specimens with restraint combination as illustrated in Fig. 9c, d and e, as base friction and print geometry did not provide enough restraint to induce cracking in the restraint combinations illustrated in Fig. 9a and b.

4.3. Shrinkage crack prediction models

Shrinkage crack prediction models are of considerable significance as they serve to mitigate the ingress of aggressive and unwanted agents [170,171]. Accurately forecasting the spaces between cracks through a crack prediction model that incorporates shrinkage strain, elastic modulus, and tensile strength could provide valuable guidance for establishing appropriate movement joint spacings of 3DCP structures, thereby minimizing the occurrence of and its associated durability and serviceability concerns. However, no predictive models have been proposed in relevant literature to address crack formation.

However, Moelich et al. [50] introduced a PSC risk model for 3DCP elements subjected to varying environmental conditions. This risk model is based on key parameters such as the evaporation rate, surface area to volume ratio, delay time, and concrete stiffness rate.

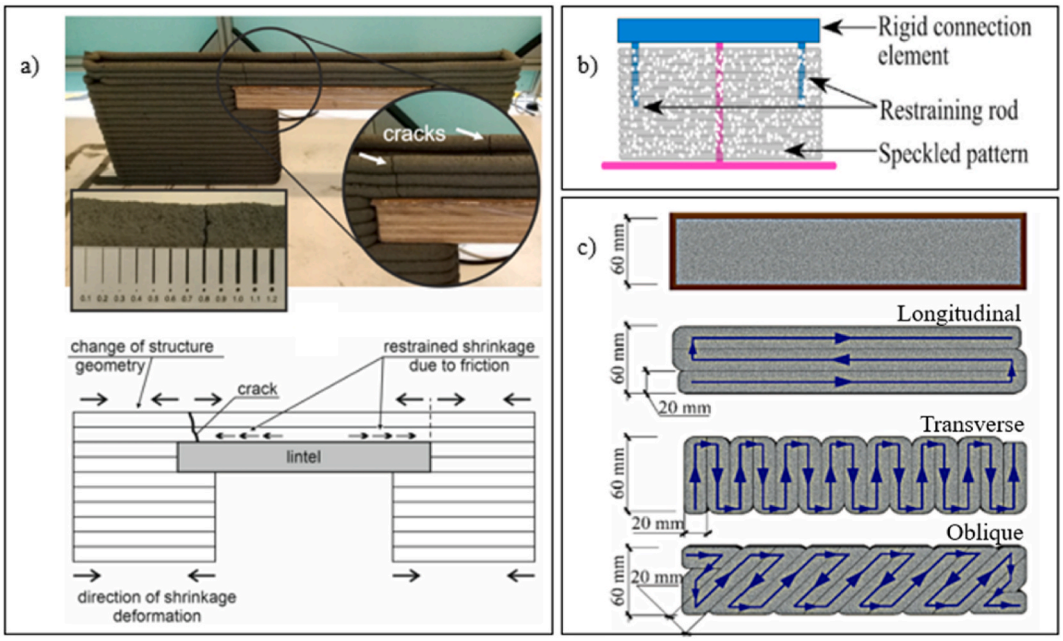


Fig. 8. Methods of inducing restrained shrinkage: a) with the use of a rigid base [169] b) with the use of steel rods [64] c) with the use of print geometry [101].

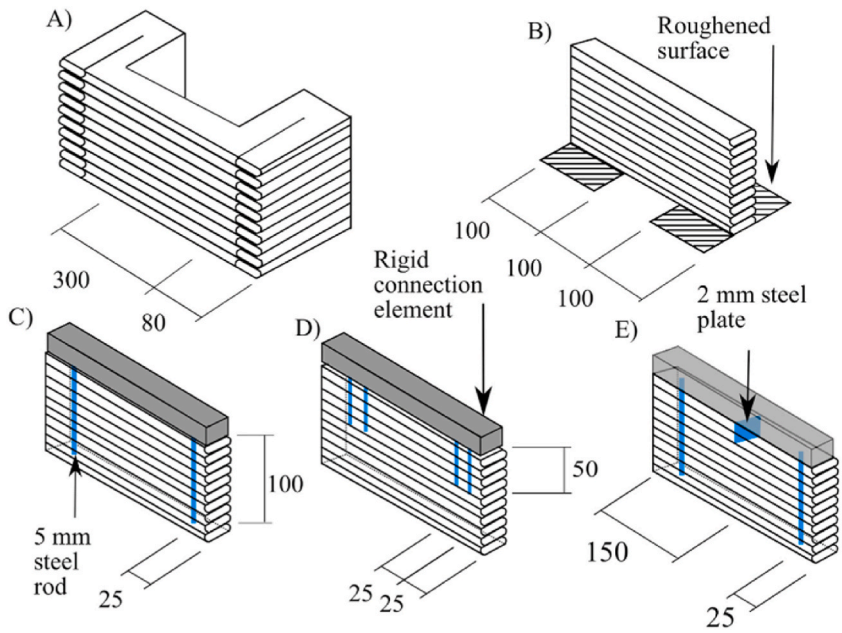


Fig. 9. **S**hrinkage restraint by: a) the printed geometry, b) a roughened base, c) inserting two steel rods connected to a rigid element, d) inserting four steel rods connected to a rigid element and e) inserting two steel rods and a notch midway between restraining rods [14].

The relationship between these parameters is expressed in Equation (9):

$$CR \propto c_1 \times EvR \times \frac{A}{V}\left(\frac{T_i + T_f}{2} - T_d\right) - c_2 \tag{9}$$

where the crack risk (CR) increases linearly with evaporation rate (EvR in $kg/m^2/h$) and the surface area to volume ratio (A/V in m^2/m^3) and CR decreases linearly with delay time (T_d in h) until between the initial (T_i in h) and final (T_f in h) setting time and $c_1 = 0.00489$ and $c_2 = 2$ are regression constants related to the mix proportions and materials.

This model defines the crack risk as the product of the normalised crack area and the crack probability. The normalised crack area is derived from the mean crack area of all tested specimens divided by the maximum crack area observed across all specimens. In contrast, the crack probability is determined by the ratio of specimens exhibiting cracking to the total number of tested specimens under specific drying conditions. In agreement with previously discussed studies, this crack risk prediction model reveals that the evaporation rate, in combination with the development of stiffness, directly influences the cracking risk of 3DPC in the plastic state.

5. Case studies

3DCP has been applied to an increasing number of projects undertaken worldwide [3]. To date, 3DCP has been used to construct houses, apartment buildings, urban furniture, pedestrian bridges and office buildings [172–174]. Some of the mentioned projects are shown in Fig. 11 [175–177].

If the structures depicted in Fig. 10 are printed on-site, they are immediately exposed to harsh ambient conditions during and directly post-printing. The reviewed literature shows that these harsh conditions may exacerbate shrinkage and increase the risk of cracking. Therefore, to mitigate the influence of unfavourable environmental conditions, the bicycle bridge in Fig. 10a was constructed at an off-site location where printing takes place in a controlled environment. Instances of initial and long-term shrinkage cracking have been reported, as illustrated in Fig. 11, where cracking is present in printed structures in Europe [59,60,178]. The severity of shrinkage cracking is evident in Fig. 11, where cracks propagate from the bottom to the top of the wall section. These cracks also propagate through the thickness of the wall section, compromising printed structures' aesthetics, durability, and serviceability. 3DPC standards under development should address the shrinkage-related challenges of concrete printing by providing a guideline for movement joint spacings to avoid cracking.

It must be noted that reinforcement significantly influences apparent shrinkage and cracking. The case studies shown in Fig. 10 do not include typical conventional reinforcement in the 3DPC, but are respectively unreinforced 3DPC elements that are post-tensioned to form a bridge span (Fig. 10a), and 3DPC that acts as stay-in-place formwork into which reinforced concrete was cast in both the Dubai offices and Beckum house (Fig. 10b and c), and unreinforced 3DPC in the Milestone house (Fig. 11).



Fig. 10. a) 3DCP bicycle bridge [139], b) 3DCP administrative building in Dubai [138] and c.1) shows the exterior and c.2) the interior of the first 3DCP building constructed in Germany [177].

6. Conclusions

This comprehensive review of the existing literature indicates that shrinkage remains a significant challenge to the long-term performance and structural integrity of 3DPC. The distinctive characteristics of 3DPC like the absence of formwork, early-age exposure and mix design constraints contribute to elevated shrinkage rates, increased risk of cracking and the subsequent durability concerns,

To address these issues, future research should not only understand the mechanisms of shrinkage, but also develop and implement practical solutions. In conclusion, the following areas are pivotal:

- Given the inconsistencies reported in shrinkage values as a result of the various test methodologies, zero-time, environmental conditions and sample sizes there is a need for a standardised test method to measure the plastic and drying shrinkage of 3DPC where the influential factors need to remain constant. This will allow for direct comparison of shrinkage results.
- 3DPC exhibits significantly larger shrinkage strains in comparison to cast concrete, due to the inherent differences in mix design, like reduced w/c ratios, higher paste content and reduced aggregate content. Tailored SRAs, internal curing agents and fiber reinforcements should be explored to limit drying and autogenous shrinkage of 3DPC. Similarly, plastic shrinkage and the consequent cracking is mitigated by the incorporation of short polypropylene microfibers.
- The absence of formwork exposes the concrete to environmental conditions directly after extrusion. Mitigation measures are required through the use of moisture-retaining covers or the application of curing compounds to control and reduce moisture loss and the resulting early-age shrinkage deformations.
- Lastly, developing shrinkage and crack-prediction models specific to 3DPC is necessary. These models should account for environmental conditions, the time dependent mechanical properties, interlayer pass time and the geometric restraints. Once tested and validated these models could serve as a foundation for designing movement joint spacings, printing strategies and reinforcement placement to control restrained shrinkage cracking.

In conclusion, shrinkage of 3DPC is mitigated by interactively combining material innovations, environmental control, predictive modelling and testing standardization to produce durable, shrinkage resistant 3DPC.

CRediT authorship contribution statement

Heinrich Els: Writing – original draft, Methodology, Conceptualization. **Gideon van Zijl:** Writing – review & editing, Supervision, Conceptualization. **Wibke de Villiers:** Writing – review & editing, Supervision, Conceptualization.

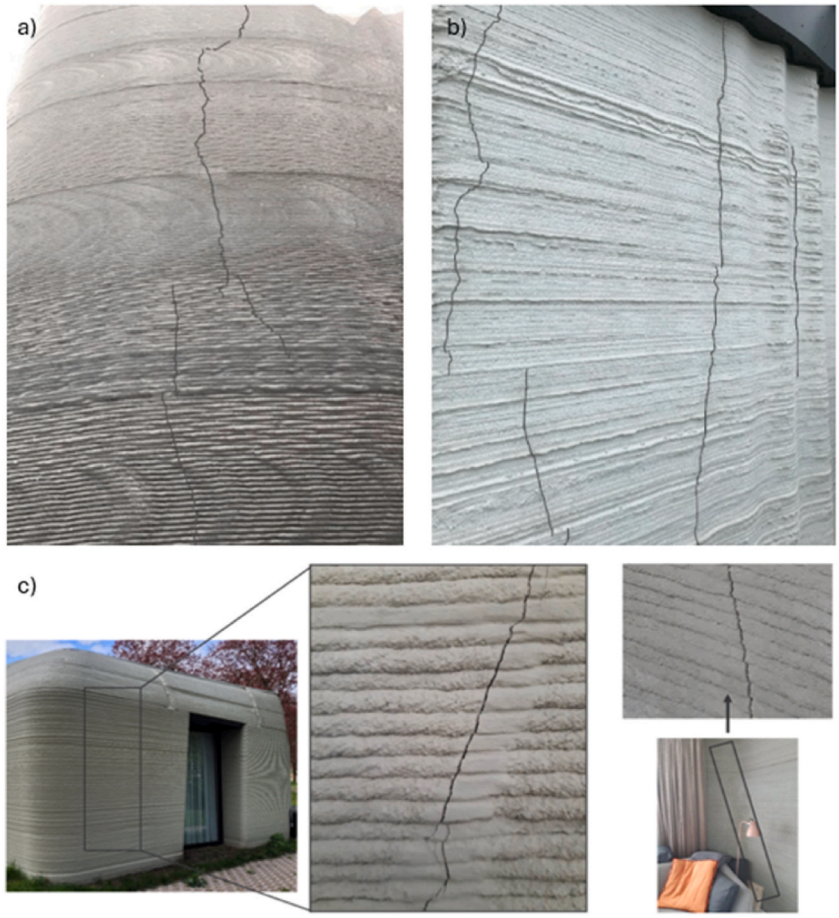


Fig. 11. Shrinkage-related cracking present in a) a two-story residential building printed by C. Kamp [178], b) the first 3DPC structure in Europe by Markin et al. [49] and c) the outer and inner walls of a residential building constructed in the Netherlands [56]. Note: these images are reported without scale, and the cracks in a) and b) are highlighted in black.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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